

First Hop Redundancy Protocol (FHRP)

First Hop Redundancy Protocol (FHRP) is a family of networking protocols designed to protect the default gateway in a subnetwork by allowing multiple routers to share a virtual IP address and virtual MAC address. This configuration ensures high availability and transparent failover, meaning if the active router fails, a standby router automatically takes over within seconds without disrupting network connectivity.

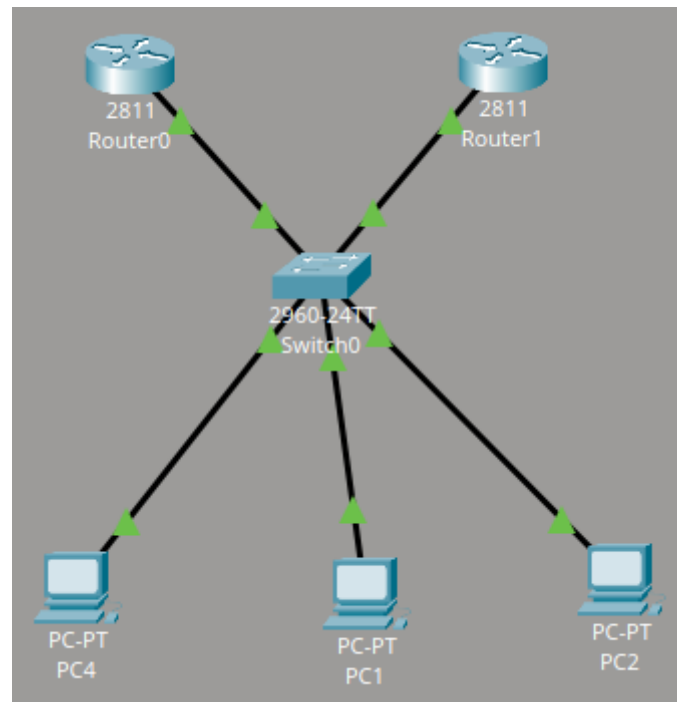
The most common FHRP implementations include:

- 1) HSRP (Hot Standby Router Protocol): A Cisco-proprietary standard developed in 1994 that provides basic redundancy with one active and one standby router.
- 2) VRRP (Virtual Router Redundancy Protocol): An open standard (IETF) based largely on HSRP, allowing for multi-vendor interoperability.
- 3) GLBP (Gateway Load Balancing Protocol): A Cisco-proprietary protocol that, like HSRP and VRRP, provides redundancy but adds load balancing capabilities across multiple routers.

These protocols operate by having routers in a group communicate via periodic messages; if the active router stops sending these messages, the standby router detects the failure and assumes the role of the active gateway, ensuring zero downtime for end-users.

- 1) `standby 10 ip 10.10.10.1` Enables the **Hot Standby Router Protocol (HSRP)** for standby group **10** and assigns the virtual IP address **10.10.10.1**. Devices on the network (like PC1) will use this virtual IP as their default gateway.
- 2) `standby 10 priority 110` Sets the router's HSRP priority for group 10 to **110** (default priority is 100). The router with the highest priority becomes the Active router that handles traffic. In this topology, R1 will act as the Active router, while R2 (at default priority 100) acts as the Standby router.
- 3) `standby 10 preempt` Allows this router to immediately take over as the Active router if its priority becomes higher than the current Active router (e.g., when R1 reboots after an outage). Without this command, R2 would remain Active even after R1 comes back online.

Lab : design FHRP with DHCP



router 0 : primery

```
! Configure the interface
interface GigabitEthernet0/0
ip address 10.10.10.2 255.255.255.0
no shutdown
standby 10 ip 10.10.10.1
standby 10 priority 110
standby 10 preempt
exit
```

```
! Configure DHCP Pool (exclude fixed IPs first)
ip dhcp excluded-address 10.10.10.1 10.10.10.10
```

```
ip dhcp pool LAN_POOL
network 10.10.10.0 255.255.255.0
default-router 10.10.10.1
dns-server 8.8.8.8
exit
```

```
router 1 : secondary
! Configure the interface
interface GigabitEthernet0/0
ip address 10.10.10.3 255.255.255.0
no shutdown
standby 10 ip 10.10.10.1
standby 10 priority 100
standby 10 preempt
exit
```

```
! Configure Backup DHCP Pool (same subnet and default-router)
ip dhcp excluded-address 10.10.10.1 10.10.10.10
```

```
ip dhcp pool LAN_POOL
network 10.10.10.0 255.255.255.0
default-router 10.10.10.1
dns-server 8.8.8.8
exit
```

Verification Commands

- `show standby brief` — Verify which router is **Active** and which is **Standby**.